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Submitter Information

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General Comment

Thank you very much for giving me the opportunity to provide input to this important Action Plan.

My input paper is attached as a PDF file.

Attachments

Development of an Artificial Intelligence Action Plan_Input_(Final)

Development of an Artificial Intelligence Action Plan_Input_(14Mar25)

Development of an Artificial Intelligence (AI) Action Plan: Input

By Gordon Alfred Long

Executive Summary: Artificial Intelligence (AI) technology can help solve problems and mitigate challenges presented by today's rapidly-developing infrastructure, transportation, and communications networks. Core functional components of AI such as process monitoring and control, machine learning, human speech understanding, pattern recognition, and many other essential capabilities as well, can seamlessly and cost-efficiently compliment human abilities and perform complex tasks which are not possible without computer assistance. However, few (if any) of these immediately useful AI applications require extremely large (i.e., expensive) data processing centers or human-like automated simulacra. Instead of questionable attempts to replicate human forms and behavior on a grand scale that one sees throughout today's news, it would seem advisable for the U.S. Government initially to seek solutions to specific problems that AI can actually solve right now, and gradually expand the scope of these simpler AI-based solutions systematically over time.

The purpose of this input to the AI Action Plan is to recommend some of the means by which targeted, straightforward application of AI to known technical problem areas can produce useful results quickly and in cost-effective ways, which in turn may be considered suitable for inclusion in the soon-to-be-published OSTP and NITRD NCO Action Plan.

Background: AI practitioners have developed complex lexicons to describe current subdivisions of the field, but at the most basic level, there are two types of AI entity, rule-based and self-taught. There can also be hybrids of these two types of AI architectures as well.

In a rule-based AI system, the rules that are imposed on the embedded AI entity govern system functionality, and the AI has little to no discretion about how it performs. An example of this type of AI (and there are many such) is an inventory control system for a business or warehouse. The rules written into the AI's software are followed explicitly, and if additional functions or different behavior is desired, the rules must be manually updated to reflect the intended new behavior of the system. The principal advantage of such a system is that the governing ruleset is well understood and precisely defined, so that correct system behavior is reliable and unsurprising. Malfunctions are easily diagnosed and can be corrected in a straightforward manner. The principal disadvantage of this type of AI entity is that developing and maintaining the fixed ruleset is usually quite time consuming (and thus costly) and often certification of new software versions can require numerous management-level approvals, which frequently introduces further administrative complexity and potential delays.

In contrast, an AI entity which is designed to learn on its own or teach itself essentially develops its own ruleset, and its success is judged on the basis of the correctness of its performance. A useful example from several decades ago is a photograph or reconnaissance image interpretation system which was originally intended to detect camouflaged armored vehicles concealed in treelines on the forested edges of open terrain. System developers would

“educate” this system by presenting large numbers of images to the embedded AI entity and “asking” it to point out and (if possible) identify any camouflaged vehicles it detects. The developers would then score the results and feed the correct and incorrect machine answers back into the machine. Then a new set of images would be presented to the machine, scored, and fed back. This process was repeated many times, with observed increases in accuracy duly noted. Eventually the “skill level” of the AI entity would be pronounced acceptable, or if the learning process was judged to be too slow, either the AI’s sensors or its image discrimination software would be modified. The principal advantage of this methodology is that there is no need to develop complex rulesets, or maintain them – the AI entity itself does that work. And, as it processes more and more images (or other types of data) which are in turn scored and fed back, its proficiency steadily improves. The principal disadvantage is that developers cannot know exactly what rules the AI entity has created for itself, which sometimes may cause uncertain outcomes or make malfunction diagnosis difficult. However, when system analytical tasks are complex or data volumes are high – or if immediate warning of undesirable events and mishaps is needed -- this type of AI entity is generally preferred.

Strongly recommend president Trump’s new AI policy – and the Action Plan – focus initially on how best to add hybrid combinations of these two types of systems to existing hardware and software architectures to provide supervisory and monitoring functionality that humans alone cannot perform. Certified expert human oversight and approval should of course be mandatory for any actions taken in response to an AI entity’s outputs or recommendations.

Cost-effective First Applications: In view of the rapid development of cyber criminals’ hacking techniques and the vast improvement seen lately in their skill levels – especially when nation states are involved – adding AI system monitoring entities and their supporting hardware and software to critical electronic systems should be assigned very high priorities. Unassisted human capabilities are inadequate to monitor and manage complex mission-essential computer systems and networks, and/or to process very large amounts of data in near-real time. These so-called “Big Data” challenges present fertile ground for achieving quick and profitable Returns on Investment (ROI) after the necessary AI installations are accomplished. Development of a body of common knowledge and practice should also be called for in the Action Plan to enable efficient creation of standardized, distributed, cost-effective self-teaching AI entities. The driving concept would be to create widely-distributed AI “Systems of Systems” that employ largely undetectable monitoring capabilities to observe systems operations in detail, and incorporate machine learning, human speech understanding and interactive discussion capabilities to act as “24/7 colleagues” to assist human security managers. Ideally, such AI colleagues would be designed to be transparent to and compatible with standard firewalls and network intrusion protections, which -- although still essential for effective security operations - - inevitably lag in their ability to detect new or exotic threats. Relevant automatically updated data subscription and geospatial information services would also be provided for rule-based situational awareness and reference purposes. Each of these AI entities would be “taught” by installing it directly on the system to be monitored in a passive observation mode, immersing it

in day-to-day systems operations for several weeks, and instructing it to report periodically any system behavior that it categorizes as abnormal (with recorded data reports to substantiate its “judgement”). The assigned human security official would then interactively discuss the strengths and weaknesses of the AI entity’s findings, with explanations why the observed unusual behavior was or was not within normal performance expectations, especially emphasizing how much risk to system security might be involved. As the AI entity’s performance improves, the human security officer working with it would naturally develop more and more trust in the AI entity, and could (with appropriate higher-level organizational approval) increase the scope of what functions it performs. Eventually the human-AI team could develop reliable systems of notification and warning whereby appropriate officials could be immediately informed of malicious events or system malfunctions which would otherwise not be reported in a timely manner, or may even not be noticed at all. Such a set of hybrid capabilities would be especially valuable in detecting and countering Advanced Persistent Threats (APTs).

In similar fashion, AI entities could be set to monitor aircraft flight control and avionics systems continuously to detect and mitigate potentially damaging or fatal malfunctions. These AI-powered monitoring systems could also be programmed to detect and report malware secretly installed in aircraft on-board systems by hostile forces. Costs involved with installing such AI monitoring systems are likely not to be prohibitive, especially when compared to the costs associated with recovery and cleanup from the adverse events they would be able to warn of and protect against.

But perhaps the most effective way in which AI could be used immediately is to improve the reliability of long-haul and wide-area networks, which routinely suffer from extensive outages and dropped messages, incomplete reconfiguration plans, and general lack of in-network malfunction diagnostics. Personal direct experience using very capable network performance diagnostic tools soon revealed that these tools had no capability at all to fix the problems they discovered, or even reroute message traffic around problem nodes or links. An AI entity could assist greatly in providing and managing appropriate corrective capabilities for typical network outages and degradations.

In addition to near-term cost-benefit advantages to be achieved by initiating Action Plan work with relatively less-ambitious AI projects for these and many similar applications, there will arise naturally-occurring concurrent protection methods (discovered through hands-on engineering experience) against the dangers which many fear are possible were a complex AI entity to run amok. With considerable numbers of such focused AI projects in progress concurrently, much will also be learned collectively about what does and does not work, and collaborative “best of breed” production decisions will be enabled.

Recommendations: The principal recommendation this paper presents is to *assign at least half of the projects the Action Plan will direct to developing immediate or near-term AI applications which are specific to known problems and technical challenges and can be implemented with existing technology in cost effective ways.* All three possible AI applications described above exhibit these characteristics, and technical means for implementing the basic functions involved have been generally known (but typically only applied piecemeal) for at least a decade and a half. The most urgent needs now are to integrate these critical supporting functions into AI entities which can serve as respected colleagues for human operators, rather than as slaves or mere servants – and to compile a supporting Common Body of Knowledge.

The second recommendation is to *approach testing of AI solutions in much the same ways that Special Government Employee Elon Musk (in his civilian role as SpaceX CEO) tests spacecraft hardware and software – in actual U.S. government and civilian space systems that are either in regular use or in development.* In the case of AI then, test new designs directly on existing systems and platforms which are known to be in need of continuous 24/7 monitoring, or will perform better with near-real-time problem identification/reporting, and will benefit from effective AI corrective actions and proposals for improvement.

Along with implementation of these simple recommendations will come significant aggregate performance improvement in operational and developmental systems to which AI is applied, in much the same fashion as occurs naturally in self-teaching AI systems themselves.

Respectfully submitted by: Gordon Alfred Long

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